

1 WHERE – Filtering (10 Questions)

1. Retrieve all transactions where the payment method is Credit Card.
 2. Find items where the fat content is “Low Fat” and the brand is “ITC”.
 3. Get all outlets established before the year 2000.
 4. Show transactions where the unit price is greater than 200.
 5. Display all outlets located in Tier 3 cities with competitors present.
 6. Find all transactions where the units sold are more than 3 and the unit margin is less than 50.
 7. List all items belonging to Soft Drinks or Frozen Foods.
 8. Show all outlets whose average income level is between 40,000 and 55,000.
 9. Retrieve all transactions made in the year 2023.
 10. Find all items whose names contain the word “Energy”.
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2 GROUP BY – Aggregations (10 Questions)

1. Find the total units sold per item.
 2. Show the average unit price by brand.
 3. Count the number of transactions per payment method.
 4. Calculate the total revenue per outlet.
 5. Show the average unit margin for each item type.
 6. Find the number of outlets in each location type.
 7. Display the average income level grouped by outlet size.
 8. Show total units sold per brand across all outlets.
 9. Find the average customer age and household size for each outlet type.
 10. Calculate the total profit per year.
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3 ORDER BY – Sorting (10 Questions)

1. List all items ordered by brand name alphabetically.
2. Display transactions sorted by transaction date (most recent first).

3. Show outlets sorted by average income level descending.
 4. Retrieve all items sorted by item type and then by item name.
 5. List the top-selling items by units sold in descending order.
 6. Show outlets ordered by distance to transportation hubs (closest first).
 7. Sort transactions by unit margin descending.
 8. Display brands ordered by their average unit price (high to low).
 9. List all outlets ordered by establishment year (oldest first).
 10. Sort transactions by payment method and then by unit price.
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 LIMIT – Restricting Results (10 Questions)

1. Display the first 5 items from the items table.
 2. Show the top 10 transactions with the highest unit margin.
 3. Retrieve the 5 most recently established outlets.
 4. List the 3 most expensive items based on unit price.
 5. Show the top 5 outlets with the highest average income level.
 6. Display the top 10 transactions where units sold is highest.
 7. Retrieve the 5 oldest outlets.
 8. Show top 5 brands with the most items.
 9. List top 10 profitable transactions by total profit.
 10. Show 5 outlets farthest from transportation hubs.
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 WINDOW FUNCTIONS (10 Questions)

1. Rank items by total units sold.
2. Find the top outlet by revenue within each outlet type.
3. Show running total of sales per outlet.
4. Compute average margin per brand using a window function.
5. Number each outlet by establishment year order.
6. Find the difference between each transaction's margin and average margin.

7. Show each outlet's highest and lowest transaction price.
 8. Rank brands by average price using DENSE_RANK.
 9. Calculate cumulative units sold per date across all outlets.
 10. Compute each outlet's sales percentage of total sales.
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6 CUMULATIVE & AGGREGATE FUNCTIONS (10 Questions)

1. Find the total units sold across all transactions.
 2. Get the average, maximum, and minimum unit prices.
 3. Count the number of unique customers.
 4. Calculate total profit by item type.
 5. Find average income level for each outlet type.
 6. Count transactions by payment method.
 7. Find maximum and minimum customer age per outlet size.
 8. Calculate total margin by brand.
 9. Get yearly total units sold and average unit price.
 10. Find total and average income per outlet location type.
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7 CTE – Common Table Expressions (10 Questions)

1. Create a CTE to calculate total revenue per outlet.
2. Use a CTE to find average unit margin per item type.
3. Create a CTE for high-value transactions (above ₹500 total).
4. Use a CTE to find outlets with above-average income.
5. Build a CTE joining all tables to compute total profit per brand.
6. Write a CTE to get the top 5 outlets by revenue.
7. Create a CTE for total yearly profit and outlet count.
8. Use a CTE to list outlets in Tier 3 with competitors present.
9. Create a CTE to find average units sold per outlet.
10. Write nested CTEs to find the top 5 brand–outlet revenue combinations.

